

# MyoTrack

*A Modular Wearable sEMG System for Real-Time Muscle*

*Activation Monitoring and Repetition Counting*

## TEAM MEMBERS

Dane Czajkowski

Cristian Frutos

Robert Lakatos

Ian Smith

Will Gillice

## BACKGROUND

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Strength-training athletes and physical therapy patients have no accessible way to see which muscles are actually firing during an exercise, or to objectively confirm how many quality repetitions were completed. Existing surface EMG (sEMG) systems are typically clinical-grade, expensive, wired, and impractical for everyday gym use.

A low-cost, wireless, multi-channel wearable EMG platform would let users place sensors on any muscle group, view activation live, and automatically log workout data — closing the gap between clinical biosignal tools and consumer fitness wearables. The goal of this project is to design and build a small set of battery-powered, wireless sEMG modules paired with a companion application that visualizes muscle activity in real time and automatically detects and counts repetitions.

## PROJECT NEEDS

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1. Acquire surface EMG signals from skin-contact electrodes with sufficient gain and noise rejection to resolve muscle activation.
2. Filter and condition the analog signal to remove motion artifacts.
3. Digitize and process the EMG signal on a low-power microcontroller in real time.
4. Wirelessly transmit processed activation data from each module to a host device with low latency.
5. Support multiple simultaneous wearable modules without data collisions or drift between channels.
6. Run for the duration of a typical workout session on a rechargeable battery.
7. Provide a companion application that displays live activation levels per muscle and logs session data.

## PROJECT SCOPE

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- Analog front-end design: instrumentation amplifier, band-pass filtering, and DC bias circuitry for sEMG acquisition
- PCB design and layout for each wearable sensor module
- Embedded firmware for signal sampling, basic on-device filtering, and radio communication
- Wireless communication protocol design (e.g., BLE) supporting multiple concurrent modules
- Battery management and low-power operation for wearable form factor
- Digital signal processing for envelope detection and repetition-detection algorithm development

- Companion mobile application development for live visualization, muscle selection, and data logging
- Bench and on-body testing to validate signal quality, wireless reliability, and repetition-count accuracy

## **DELIVERABLES**

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1. At least two functional wearable sEMG sensor modules with wireless transmission.
2. A companion application showing real-time multi-channel muscle activation.
3. A working repetition-detection algorithm validated against manually counted reps for several exercises.
4. Recorded and reviewable workout session data within the application.
5. Test results characterizing signal quality (noise floor, artifact rejection) and wireless reliability/range.
6. Final documentation: schematics, PCB files, firmware source, and a user/build guide.
7. Live demonstration of two or more modules operating simultaneously during a sample workout.

## **ASSUMPTIONS AND DEPENDENCIES**

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- Assumes a single target muscle group and exercise type (e.g., bicep curls) for primary validation, rather than general-purpose use across all lifts.
- Fatigue-index accuracy is expected to be validated relatively, not as a clinically precise measurement.
- Assumes reasonably consistent electrode-skin contact during motion; a fully motion-proof electrode system is not required.
- Battery life and wireless range are scoped to a single workout session in a normal gym environment.
- Multi-muscle or multi-sleeve fatigue comparison is a stretch goal, not required for core project success.
- Assumes access to standard EMG electrodes and conductive fabric or gel-based contacts; custom electrode/textile fabrication is out of scope beyond basic integration.